

Dataset Link : [Data](#)

Problem 1

Tasks to be done after loading the data

A. Data preprocessing

1. Check missing values.
2. Identify numerical and categorical features.
3. Apply appropriate **encoding** to categorical columns.
4. Check whether numerical features have very different scales.
5. Apply appropriate **feature scaling**.

B. Skewness

6. Calculate skewness for all numerical variables.
7. Identify highly skewed features.
8. Apply an appropriate transformation to reduce skewness.
9. Compare the distribution before and after transformation using distribution plots.

C. Feature Selection

10. Determine which features are useful for customer segmentation.
11. Remove irrelevant/redundant features and explain your reasoning.

D. Clustering

12. Apply **K-Means** clustering.
13. Use the **Elbow Method** to determine an appropriate K.
14. Calculate the **Silhouette Score** for different K values.
15. Select the final number of clusters.

Problem 2

Use the same customer dataset given above.

This time **do NOT use K-Means as your final algorithm.**

Tasks

1. Preprocess the data.
2. Encode categorical variables.
3. Scale the numerical features.
4. Perform feature selection.
5. Apply **Agglomerative Hierarchical Clustering**.
6. Experiment with different linkage methods:
 - Ward
 - Complete
 - Average
7. Generate a **dendrogram**.
8. Decide the number of clusters from the dendrogram.
9. Fit Agglomerative Clustering with your chosen number of clusters.
10. Calculate the Silhouette Score.

Problem 3

Use the same customer dataset given above.

id	age	gender	income	spending_score	membership_years	purchase_frequency	preferred_category	last_purchase_amount
----	-----	--------	--------	----------------	------------------	--------------------	--------------------	----------------------

There are **9 features**

Problem

You want to visualize the customers in 2D, but 9 dimensions are difficult to visualize.

Tasks

1. Check missing values.
2. Handle categorical variables if present.
3. Check skewness.
4. Apply transformations where necessary.
5. Scale the features.
6. Perform feature selection.
7. Apply **PCA**.
8. Calculate explained variance.
9. Plot:
 - Explained variance per component
 - Cumulative explained variance
10. Determine how many components are required to explain at least **90% variance**.
11. Transform the original dataset using those components.
12. Plot the first two principal components.

Dataset link : [Data 2](#)

Problem 4

Tasks

1. Convert the transaction data into the appropriate format for Apriori.
2. Calculate item frequencies.
3. Apply the **Apriori algorithm**.
4. Try: min_support = 0.01
5. Generate frequent itemsets.
6. Generate association rules.
7. Calculate: Support Confidence Lift
8. Sort the rules by lift.
9. Find the top 10 strongest rules.